Hints: Number theory – 4

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Every prime $p \in (m+1, 2m+1]$ divides $\binom{2m+1}{m} \leqslant 4^m$. So $\prod_{m+1<p\leqslant 2m+1} p \leqslant 4^m$. Induct: even n is not prime, and for odd $n = 2m+1$ use the bound up to $m+1$.

Hint 2. Bertrand: suppose there is no prime in $(n, 2n]$. By Kummer, $p^{v_p} \leqslant 2n$ for every p, so primes $p > \sqrt{2n}$ appear at most to the first power, and primes in $(2n/3, n]$ do not divide $\binom{2n}{n}$ at all.

Hint 3. Then $\frac{4^n}{2n+1} \leqslant \binom{2n}{n} \leqslant (2n)^{\sqrt{2n}} \cdot 4^{2n/3}$. This fails for n large (about $n \geqslant 500$). For smaller n, use the chain of primes $2, 3, 5, 7, 13, 23, 43, 83, 163, 317, 631$.

Problem 2

Note. As printed: (a) $\operatorname{lcm}(1, \dots, n) = e^{(1+o(1))n}$ is equivalent to the prime number theorem and is not elementary. (b) $\binom{2n}{n}$ is *not* always divisible by $\operatorname{lcm}(1, \dots, n+1)/(n+1)$ ($n = 3$: 20 vs. 3). The correct statements are that $\binom{2n}{n}$ divides $\operatorname{lcm}(1, \dots, 2n)$, and that $\binom{n}{k} \mid \operatorname{lcm}(1, \dots, n+1)/(n+1)$. The hints prove Chebyshev-type bounds and the correct divisibility.

Hint 1. By Kummer, $v_p\binom{2n}{n} \leqslant \#\{j : p^j \leqslant 2n\} = v_p(\operatorname{lcm}(1, \dots, 2n))$, so $\binom{2n}{n} \mid \operatorname{lcm}(1, \dots, 2n)$.

Hint 2. Hence $\operatorname{lcm}(1, \dots, 2n) \geqslant \binom{2n}{n} \geqslant \frac{4^n}{2n+1}$. Together with $\operatorname{lcm}(1, \dots, N) \leqslant N^{\pi(N)}$ this gives $\pi(2n) \geqslant \frac{n \log 4 - \log(2n+1)}{\log 2n}$.

Hint 3. Upper bound: $\operatorname{lcm}(1, \dots, N) = \prod_p p^{\lfloor \log_p N \rfloor} \leqslant \prod_{p\leqslant N} p \cdot \prod_{p\leqslant\sqrt{N}} N \leqslant 4^N N^{\sqrt{N}}$. So $e^{cN} \leqslant \operatorname{lcm}(1..N) \leqslant e^{CN}$ elementarily. The constant 1 in the exponent needs the PNT.

Problem 3

Hint 1. $v_2\binom{2n}{n} = s_2(n)$ (Kummer). So $v_2 = 1$ exactly when n is a power of 2.

Hint 2. Square-free forces $v_2 \leqslant 1$, hence $n = 2^k$. Check small cases: $n = 1, 2, 4$ give $2, 6, 70$, but $n = 8$ gives $12870 = 2 \cdot 3^2 \cdots$.

Hint 3. That no other n works is a deep theorem (Erdős's conjecture, proved by Sárközy for large n and completely by Granville–Ramaré). Kummer for $p = 3, 5, \dots$ shows why: one needs two carries in some base. Answer: $n \in \{1, 2, 4\}$.

Problem 4

Note. $v_p(\sum k^{-2})$ has no closed form: it equals 1 for $p \geqslant 5$, except for the Wolstenholme primes (16843, 2124679, …), where it is $\geqslant 2$. For $p = 3$ it is 0. The hints prove $v_p \geqslant 1$ and the link with H_{p-1}.

Hint 1. Pair k with $p-k$: $H_{p-1} = \frac{p}{2} \sum_k \frac{1}{k(p-k)}$, and $\frac{1}{k(p-k)} \equiv -\frac{1}{k^2} \pmod p$.

Hint 2. $\sum_{k=1}^{p-1} k^{-2} \equiv \sum k^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0 \pmod p$ for $p \geqslant 5$. This gives $v_p(H_{p-1}) \geqslant 2$.

Hint 3. More precisely, $H_{p-1} \equiv -\frac{p}{2} \sum k^{-2} \pmod{p^3}$, so $v_p(H_{p-1}) \geqslant 3$ if and only if $v_p(\sum k^{-2}) \geqslant 2$. Generically $v_p(\sum k^{-2}) = 1$ (it is $\equiv \frac{2}{3} p B_{p-3} \pmod{p^2}$).

Problem 5

Hint 1. Odd n: take $k = n^m$. Then $a_k = n^{n^m - m}$ is odd.

Hint 2. Even n: find k with $k \mid n^k - r$ for some small r with n^k/k just above $(n^k - r)/k$. Try $k = p \cdot 2^m$, where p is an odd prime factor of $n^{2^m} - 2^m$ (it exists, since $n^{2^m} - 2^m = 2^m(\text{odd} > 1)$).

Hint 3. By Fermat, $n^k = (n^{2^m})^p \equiv (2^m)^p \equiv 2^m \pmod p$. Since $0 < 2^m < k$, we get $a_k = \frac{n^k - 2^m}{p\,2^m}$, which is odd because $n^k/2^m$ is even.

Problem 6

Hint 1. Check that $a = 1, 3, 5$ work ($n = 1, 1, 4$). For even $a = 2^\alpha d$, pick $i < a$ with $n + i = 2^{\alpha-1} e$, e odd. Then $t(n + i + a) = 2d + e \equiv e + 2 \pmod 4$.

Hint 2. Odd $a > 8$: some $n + i = 2d$ (d odd) with $i \leqslant a - 5$. Then $t(n + i)$ and $t(n + i + 4)$ differ by 2 mod 4, while $n + a + i$ and $n + a + i + 4$ are odd and congruent mod 4. So both differences cannot vanish mod 4.

Hint 3. $a = 7$: some $n + i \in \{8k + 3, 8k + 6\}$, and both cases fail directly. Answer: $a \in \{1, 3, 5\}$.

Problem 7

Hint 1. If $n = \prod p_i^{e_i}$ has 2^k divisors, then each $e_i + 1$ is a power of 2. So n is a product of distinct numbers of the form p^{2^j} ("Fermi–Dirac primes"), exactly k of them.

Hint 2. Minimality: $f(2^k)$ is the product of the k smallest elements of $\{p^{2^j}\}$. (If a smaller unused element exists, swap it for a larger one; check that the result is still of the right shape.)

Hint 3. Hence $f(2^{k+1}) = f(2^k) \cdot (\text{the next element})$.

Problem 8

Hint 1. Bertrand: there is a prime p with $n < p \leqslant 2n$ (in fact $p < 2n$ for $n > 1$).

Hint 2. By Kummer or directly, $v_p\binom{2n}{n} = 1$ (p divides $(2n)!$ once and $n!$ not at all). So $\binom{2n}{n}$ is not a perfect power.

Problem 9

Hint 1. For a fixed prime p, let $p^e \leqslant n + 1 < p^{e+1}$. Show that $\max_k v_p\binom{n}{k} = e - v_p(n + 1)$. By Kummer it is the maximal number of carries in $k + (n - k)$.

Hint 2. Let $t = v_p(n+1)$, the number of trailing digits $p-1$ of n. Show that no carry can occur in these t lowest positions (induct from position 0). Also no carry occurs at position e or above. So there are at most $e - t$ carries. Take $k = p^e - 1 \leqslant n$: then $n - k = n + 1 - p^e$, and the addition carries at every position $t, \ldots, e - 1$. Comparing with $v_p\big(\operatorname{lcm}(1, \ldots, n + 1)/(n + 1)\big) = e - t$ proves the identity.

Hint 3. For the sum, use $\frac{1}{\binom{n}{k}} = (n + 1)\int_0^1 t^k(1 - t)^{n-k}\,dt$. Summing the geometric series under the integral gives $\sum_k \frac{(-1)^k}{\binom{n}{k}} = \frac{(n+1)(1+(-1)^n)}{n+2}$.

Problem 10

Hint 1. Legendre: $v_p\Big(\frac{n!}{k_1! \cdots k_m!}\Big) = \frac{\sum s_p(k_i) - s_p(n)}{p-1}$.

Hint 2. Adding $k_1, \ldots, k_m$ column by column, each carry of size c (an amount c carried to the next position) lowers the total digit sum by $c(p - 1)$. So the valuation is the total amount carried.

Hint 3. The multinomial coefficient is coprime to p exactly when, in every base-p position, the digits of the k_i sum to the digit of n (no carries).